Session 26: Equity-Settled Share-Based Payments II: Case Study, eBay's Stock Option Plans

By the end of this session you must be able to apply Session 25's IFRS2 framework to a real pre-IFRS2 accounting choice (intrinsic value vs. fair value expensing of stock options), interpret eBay's actual footnote disclosures, and discuss why the choice between the two methods can flip a reported profit into a reported loss.

Plan

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1. The Case Hook: Was eBay Profitable or Not?

- eBay reported net income of $48 million for 2000. But under a then-new accounting disclosure requirement, the company also had to disclose what its earnings would have been had it expensed stock options at their estimated fair value: a **loss** of almost $91 million.
- The case opens on this striking contrast, the same year, the same company, two completely different profit pictures depending purely on how one type of compensation cost is accounted for, and asks the reader to work out why.

2. eBay's Business and Its Reliance on Stock Options

- eBay, founded in 1995 and taken public in September 1998, grew from 1 employee to 1,927 by 2000, and explicitly identified its dependence on key personnel as a major risk factor in its SEC filings, noting intense competition for technical and managerial talent (especially in the San Francisco Bay Area) and that job candidates in the internet/tech industry "often consider the value of the stock options they are to receive in connection with their employment."
- The company adopted multiple stock option plans, approved by shareholders via board-recommended proxy statements, explicitly to attract, motivate and retain employees, directors and consultants.
- By year-end 2000, eBay had granted over 9 million options that year alone, with over 26 million unexercised options outstanding against roughly 260 million shares outstanding. A substantial proportion of the company's equity was, in effect, promised out to employees via options.

3. Two Accounting Methods: APB25 Intrinsic Value vs. SFAS123 Fair Value

APB25 intrinsic value method (the method eBay actually used for *recognition*): compensation expense is based on the difference, measured at grant date, between the fair value of the company's stock and the option's exercise price.

SFAS123 fair value method (used only for *pro forma disclosure*, not recognition, in eBay's case): compensation cost is measured using an option-pricing model (Black-Scholes) that captures the option's full fair value, including its time value and the effect of expected volatility. Not just the immediate spread between stock price and exercise price.

- The crucial mechanical point: most of eBay's options, like the vast majority of executive options at the time, were granted *at the money*. The exercise price equal to the stock's fair value on the grant date.
- Under APB25's intrinsic value method, the "difference between fair value of stock and exercise price" at grant date is therefore **zero** for an at-the-money option, so **no compensation expense is recognised at all**, regardless of how valuable the option might become later.
- This is precisely the accounting choice IFRS2 (Session 25) was designed to close: IFRS2 requires the *full grant-date fair value* (via an option-pricing model, not just the immediate intrinsic spread) to be expensed over the vesting period, exactly matching the SFAS123 pro forma method eBay was required to disclose but not required to recognise.

4. Reading eBay's Actual Footnote Numbers

Stock option activity, 2000 (Exhibit 8): options outstanding at the beginning of 2000 = 26,236 thousand at a weighted average exercise price of $29.73, 9,037 thousand options granted during the year at a weighted average exercise price of $62.69, 4,499 thousand exercised at $6.23, 4,525 thousand cancelled at $65.41. Options outstanding at year-end = 26,249 thousand at a weighted average exercise price of $38.99. Options exercisable (vested) at year-end = 7,006 thousand at a weighted average exercise price of $27.73.

Grant-date fair value vs. exercise price: the "weighted average grant date fair value of options granted during period" for 2000 was **$103.79** per option, computed by eBay using Black-Scholes with these assumptions: risk-free rate 4.9%, expected life 3.0 years, dividend yield 0%, and (crucially) **expected volatility of 115%**: an extraordinarily high figure reflecting the dot-com-era volatility of eBay's stock. Exhibit 8 also reports a $62.69 weighted average **exercise price** for 2000 grants. These published averages do not establish that every grant was at the money or that the two measures are directly comparable.

The pro forma effect (Exhibit 8): 2000 net income "as reported" (APB25 basis) = $48,294 thousand ($48.3m). "Pro forma" net income (SFAS123 fair value basis) = a loss of $(90,677) thousand (≈−$90.7m). Basic EPS moves from $0.19 (as reported) to $(0.36) (pro forma). Diluted EPS from $0.17 to $(0.36). The same pattern, though smaller in absolute terms, appears in 1999 (as reported net income $9,567k vs. pro forma loss of $(41,357)k), while 1998 (pre-IPO, lower option volumes and a different valuation method) shows only a modest gap ($7,273k as reported vs. $6,497k pro forma).

5. Interpreting the Grant-Date Fair Value vs. Exercise Price Gap

- The case's own discussion question asks the reader to interpret why the $103.79 weighted average grant-date fair value is so much larger than the $62.69 weighted average exercise price reported for 2000 grants, and whether this makes economic sense. The averages alone do not establish comparable populations or at-the-money terms for every grant.

Key insight (Add-on, building on option-pricing intuition): Black-Scholes includes time value, so an at-the-money option can have positive fair value despite zero intrinsic value. Higher volatility and a longer life can increase that value. However, a call on a non-dividend-paying stock cannot exceed the stock price. For an at-the-money grant, the stock price equals the exercise price. Thus 115% expected volatility and a 3-year expected life do not justify a fair value above that grant's stock price or strike. The reported averages of $103.79 and $62.69 require clarification of grant terms and comparability before drawing a valuation conclusion.

- This is exactly why the intrinsic value method (APB25) and the fair value method (SFAS123/IFRS2) diverge so dramatically for high-volatility companies: intrinsic value captures none of this time value, while a proper option-pricing model captures all of it.
- The published gap cannot be validated as the expected Black-Scholes result for comparable at-the-money grants. It is an unresolved disclosure interpretation, not proof that the model produces a call value above the underlying stock price.

6. Recognition vs. Disclosure: Why It Mattered

- This case is a direct, real-world illustration of the "recognition vs. disclosure" academic literature already covered in Session 16 (which found that stock option expense specifically, Aboody 1996, is one of the settings where investors treat recognised values as more relevant than merely disclosed ones).
- eBay's fair-value option cost was fully disclosed (in the pro forma footnote), but not recognised in the primary financial statements. Meaning a reader of the income statement alone, without digging into the footnotes, would see only the $48 million profit, not the $91 million pro forma loss.
- IFRS2's move to mandatory *recognition* (Session 25) is precisely the accounting-standards response to this gap: rather than relying on footnote disclosure and hoping sophisticated users adjust for it themselves, the cost is now built directly into reported profit.

7. Assignment Questions Framework

Must know (how to structure a case answer, from the session outline's assignment questions):
• **Why compensate with options rather than cash/stock grants?** Options align employee incentives with share-price appreciation at comparatively low immediate cash cost to the company, and (under APB25) historically carried no P&L expense at all for at-the-money grants, a strong incentive-and-accounting combination, especially for a cash-constrained, high-growth company
• **Why are extensive disclosures required, and what are the disadvantages of options as compensation?** Because the accounting choice (APB25 vs. SFAS123) so dramatically changes reported profitability (Section 4), disclosure is needed for informed analysis. Disadvantages include potential dilution of existing shareholders, incentives that can become misaligned if the stock price falls far below the exercise price (noted directly in eBay's own risk-factor disclosure), and the difficulty of valuing options reliably given assumptions like expected volatility
• **Do you agree with eBay's accounting treatment (recognition vs. disclosure)?** A discussion question with no single correct answer, but the case's framing (a $48m profit vs. a $91m loss) and the recognition-vs-disclosure literature (Section 6) both point toward recognition being the more informative and less easily overlooked treatment, which is exactly the direction IFRS2 and its US equivalent (ASC718, which later replaced APB25) subsequently took
• **Interpreting $103.79 vs. $62.69:** both averages are printed in Exhibit 8. Neither their comparability nor at-the-money terms for every grant are established by these averages. The 115% expected volatility and 3-year expected life cannot explain a call value above the underlying stock price

8. Common Mistakes and Exam Traps

- Assuming APB25's intrinsic value method always produces zero expense. It produces zero expense specifically for at-the-money grants. An option granted *in the money* (exercise price below current fair value) would still generate a positive intrinsic-value expense under APB25.
- Treating the $103.79 vs. $62.69 averages as either proof of a valuation error or a validated at-the-money Black-Scholes result. Check grant terms and comparability first. Time value does not allow a call to exceed the underlying stock price.
- Confusing this case's historical US GAAP framework (APB25 vs. SFAS123, pre-dating today's US standard ASC718) with IFRS2 itself. IFRS2 was never optional-disclosure-only in the way APB25/SFAS123 coexisted. IFRS2 has always required *recognition* at fair value, which is exactly the treatment the case shows eBay avoiding via APB25.
- Missing that the case's central tension (recognition vs. disclosure) is the same theme already covered generally in Session 16's academic literature on leases. Recognise this as a recurring pattern in financial reporting research, not a one-off eBay quirk.
- Assuming the pro forma loss means eBay was not actually a successful, cash-generative business. The case explicitly notes eBay was profitable and had strong stock-price performance. The pro forma loss is an *accounting* reclassification of stock compensation cost, not a statement about the underlying business's cash economics.

9. Must Know Absolutely

Must know:
• APB25 (intrinsic value): expense = fair value of stock − exercise price at grant date, zero for at-the-money grants, which is what eBay used for recognition
• SFAS123 (fair value, Black-Scholes): full option fair value including time value, disclosed by eBay only pro forma, matching IFRS2's mandatory approach
• eBay 2000: as-reported net income $48.3m vs. pro forma (fair-value) net loss of ~$90.7m, same company, same year, opposite conclusions depending on the accounting method
• 2000 grants: published weighted average exercise price $62.69 and weighted average grant-date fair value $103.79 (Black-Scholes, 115% expected volatility, 3-year life). These averages do not establish comparability or validate an at-the-money explanation of the gap
• The case is a real-world precedent for why standard setters (IFRS2, later US ASC718) moved from optional disclosure to mandatory fair-value recognition of share-based payment cost